

Be the Agent

Play an AI agent with cards and find out what the model does, what the tools do and what carries the messages between them

An AI agent such as Claude Code gets a goal instead of a recipe. It plans its own steps, uses tools and keeps going until the job is done. In this experiment your group plays such an agent, with cards, a calculator and a few printed pages instead of a computer. One of you is the language model, one carries the messages, four of you are the tools, and one watches and keeps the log. By the end you will have run, by hand, the loop that every AI agent runs on.

Goal

Find out from the inside how an agent works: what the language model decides, what the tools do, who carries the messages between them, and why everything the model knows has to be on its stack of cards.

What you need

- A group of seven. With five or six of you, one person takes two roles (see the table below).
- The cards, printed and cut out ([be-the-agent-cards.pdf](#)): the system card, three task cards, a role card for everyone, the search index, the farm files, the web pages and plenty of blank cards.
- The Observer's sheet at the end of this document, a pen for everyone and a calculator.
- Enough room for the tools to sit apart from the Model, ideally at their own tables.
- About 30 minutes for the first two tasks.

The roles

Role	What you do	What you have
Model	You decide what happens next and write it on a card.	Only the stack of cards in front of you.
Harness	You take every card the Model writes. A tool card you check and carry to the right tool; the result you bring back and put on the Model's stack.	The role card with the list of tools.
Search	You find web pages for a search request and return their titles.	The search index: titles and keywords, but not the pages themselves.
Web	You return the full text of a page.	The web pages.
Files	You read out and store the farm's files.	The farm files and blank sheets for new ones.
Calculator	You calculate exactly what is written.	A calculator.
Observer	You keep the log: every card in order, with a note on what happened.	Everything, but you never help.

With six people, Search and Web are one person. With five, also Files and Calculator.

The rules

1. **The Model sees only its stack.** Nobody talks to it, and it never looks at the other tables. Everything it learns arrives on a card.
2. **Every card the Model writes starts with one of three words:**
 - **THINK:** a note to itself, for example **THINK: I need the size of the field first.**
 - **TOOL:** a tool call, written exactly as on the system card, for example **TOOL: read_file("fields").**
 - **ANSWER:** the answer to the task. This card ends the game.
3. **The Harness carries every card.** A **THINK:** card goes straight onto the stack. A **TOOL:** card goes to the tool it names; if no such tool exists or the call is written wrongly, the Harness writes **RESULT: error, unknown tool** itself and does not walk anywhere.
4. **A tool does exactly what the card says** and writes its answer on a new card that starts with **RESULT:.** It never repairs a call, never guesses what the Model meant and never adds anything that was not asked for. Tools do not talk to each other.
5. **Every card goes on top of the stack** in the order it was written, and it stays there.
6. **One card at a time.** The Model writes its next card only when the result of the last one is on the stack.

Steps

1. **Set up.** Hand out the role cards. The tools take their material and sit apart from the Model. The Model's stack starts with the system card, with the card for Task 1 on top of it. The Observer takes the log sheet.
2. **Play Task 1.** How many cards did it take, and how many walks did the Harness make? Did the Model write a **THINK:** card before its first tool call?
3. **Swap roles and play Task 2.** Which tools were busy, and which sat idle? Did the first search find the right page? Which value did the Model choose, and how did it decide?
4. **Count.** The Observer reads the log aloud: rounds, tool calls per tool, errors, and how many cards lie on the stack at the end.
5. **Play Task 3 if there is time.** One thing in the farm files will only be noticed by a careful Model.
6. **Compare with a real agent.** Your lecturer shows the log of a real agent session. Find the places where it thinks, calls a tool, receives a result and answers.

What you just played

In the game	In a real AI agent
the stack of cards	the context window : everything the model knows during a task
the system card	the system prompt : who the model is and which tools it has
the task card	the user's prompt , the goal
a THINK: card	the model's reasoning or plan
a TOOL: card	a tool call : the model only writes text, something else carries it out
the Harness	the harness , the program around the model that checks each call, runs the right tool and adds the result to the context
Search, Web, Files, Calculator	the tools : a search engine, a web fetcher, the file system, a code runner
a RESULT: card	the observation : the tool's result, added to the context
the Observer's log	the transcript of the session
the ANSWER: card	the stop condition : the goal is reached

The loop you ran, think, act and observe until the goal is reached, is the loop inside every AI agent. The Model never opened a file, never searched the web and never pressed a key on the calculator; it only wrote text, and everyone else in the group carried it out.

Go further

1. **Limited memory.** The stack may now hold at most eight cards. When it is full, the Model writes a **SUMMARY** : card that replaces the oldest cards; the system card and the task card always stay. Play Task 3 again. What got lost, and did it matter?
2. **Two tools at once.** The Model may now write two tool cards in one turn, and the Harness carries them in parallel. Play Task 2 again and compare the time with your first game.
3. **A new tool.** One person becomes a new tool, for example `forecast("date")` with its own page of information. Add the tool to the system card. Does the Model use it without being told to?
4. **A wrong result.** One tool secretly returns one wrong result, for example the Calculator is off by a factor of ten. Does the Model notice? What would a real agent need in order to notice?
5. **Be the tool for a chatbot.** Solve Task 2 with a chatbot in your browser. Whenever it needs a file, a web page or a calculation, you have to provide it yourself. Who is the Harness now, and who decides the next step?

Reflection

Write down your own answers first, then discuss them in your group.

1. In which round would the Model have been stuck without the tools, and which tool was used most?
2. What would have gone differently if the Model had had to give up its stack after ten cards?
3. Who did the thinking, who did the acting, and which of it could the Model actually do?

The kit

The cards are a separate PDF: [be-the-agent-cards.pdf](#). Print it on A4 and cut along the dashed lines; every sheet holds four cards the size of an index card. Pink cards belong to the Model (the system card, the three tasks and the blank cards it writes on), dark cards are the role cards, one for each player, and grey cards are the material of the tools: the search index, the farm files and the web pages. Print the last sheet with blank cards several times, because a game uses twenty or more.

All farm data and all values on the web pages are invented for this game. They are simplified teaching values and not advice for real fertilisation.

Observer's sheet

Task: _____ Model: _____ Harness: _____

Round	Card	Tool	What it said, in short	Note: error, detour, good move
1				
2				
3				
4				
5				
6				
7				
8				
9				
10				
11				
12				
13				
14				
15				
16				

Rounds: _____ Tool calls: Search ____ Web ____ Files ____ Calculator ____ Errors: _____ Cards on the stack at the end: _____